CSE4261: Neural Network and Deep Learning
Assignment-11

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Session: 2019-20

July 22, 2025

1 Face Verification: Comparison of Loss Functions

1.1 Implementation Overview

We implemented three face verification models using distinct loss functions:

- **Binary Cross-Entropy (BCE) Loss:** A Siamese network trained to predict whether a pair of images represents the same person.
- **Contrastive Loss:** A pair-based approach that minimizes the distance between embeddings of similar images and maximizes it for dissimilar ones.
- **Triplet Loss:** Trained on triplets consisting of anchor, positive, and negative images to ensure the anchor is closer to the positive than to the negative.

We preprocessed the face dataset and split it into training and validation sets. Each model was trained for 10 epochs using a similar CNN-based architecture to extract embeddings followed by fully connected layers.

1.2 Result Comparison

- **BCE Loss:**
  - Training accuracy increased steadily, reaching approximately 99.5%.
  - Validation accuracy plateaued around 78%, suggesting overfitting.
- **Contrastive Loss:**
  - Validation loss consistently decreased to 0.13, indicating good generalization.
  - The training process was relatively stable and straightforward.
- **Triplet Loss:**
  - Training loss stabilized around 0.47–0.49 with slower convergence.
  - The model required more careful triplet sampling and was harder to train effectively.

Conclusion: While the BCE-based model achieved the highest training accuracy, the model trained with contrastive loss generalized better on unseen data. The triplet loss model showed slower convergence and required more complex setup, without offering a distinct performance advantage in this scenario.

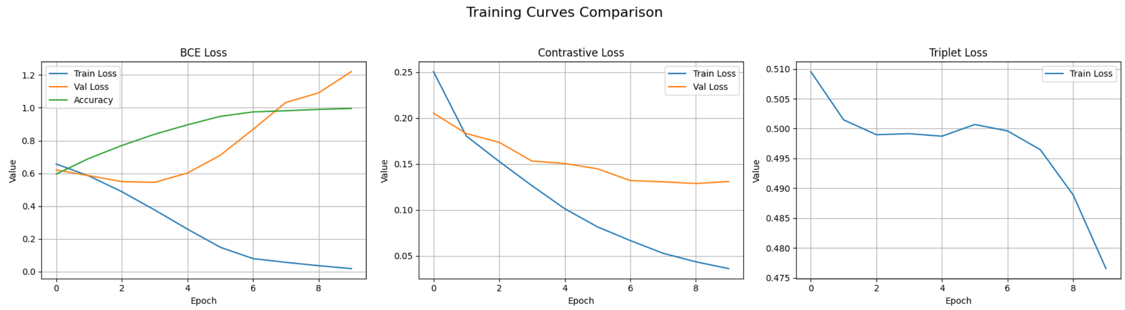
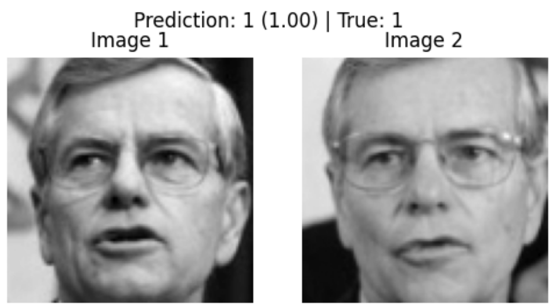
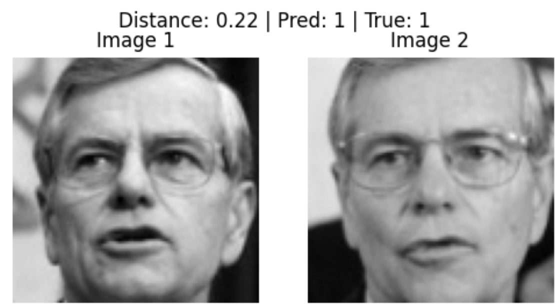


Figure 1: Training loss curves for the three face verification models: BCE-based Siamese network, contrastive loss model, and triplet loss model.



(a) Prediction using BCE-based verifier



(b) Prediction using contrastive loss verifier

Figure 2: Face verification results comparing models trained with BCE and contrastive loss.

2 VAE Training: BCE vs MSE as Reconstruction Loss

2.1 Implementation Overview

We implemented a Variational Autoencoder (VAE) on the MNIST dataset and experimented with two different reconstruction loss configurations:

- **BCE + KL Divergence:** The decoder output was trained using binary cross-entropy loss along with KL divergence for the latent space regularization.
- **MSE + KL Divergence:** The reconstruction loss was replaced with mean squared error (MSE), while KL divergence remained unchanged.

Both models were trained for 10 epochs using the same architecture and optimizer settings to ensure fair comparison.

2.2 Result Comparison

- **BCE Loss:**
  - The loss started at 169.53 and reduced to 101.67 by the final epoch.
  - The model exhibited faster convergence and generated sharper, more detailed digit reconstructions.
  - More responsive to fine-grained pixel differences.
- **MSE Loss:**

- Initial loss was 48.49, decreasing to 29.49.
- Reconstructed digits were visually smoother but appeared more blurry compared to the BCE version.
- Training was less stable with occasional visual degradation.

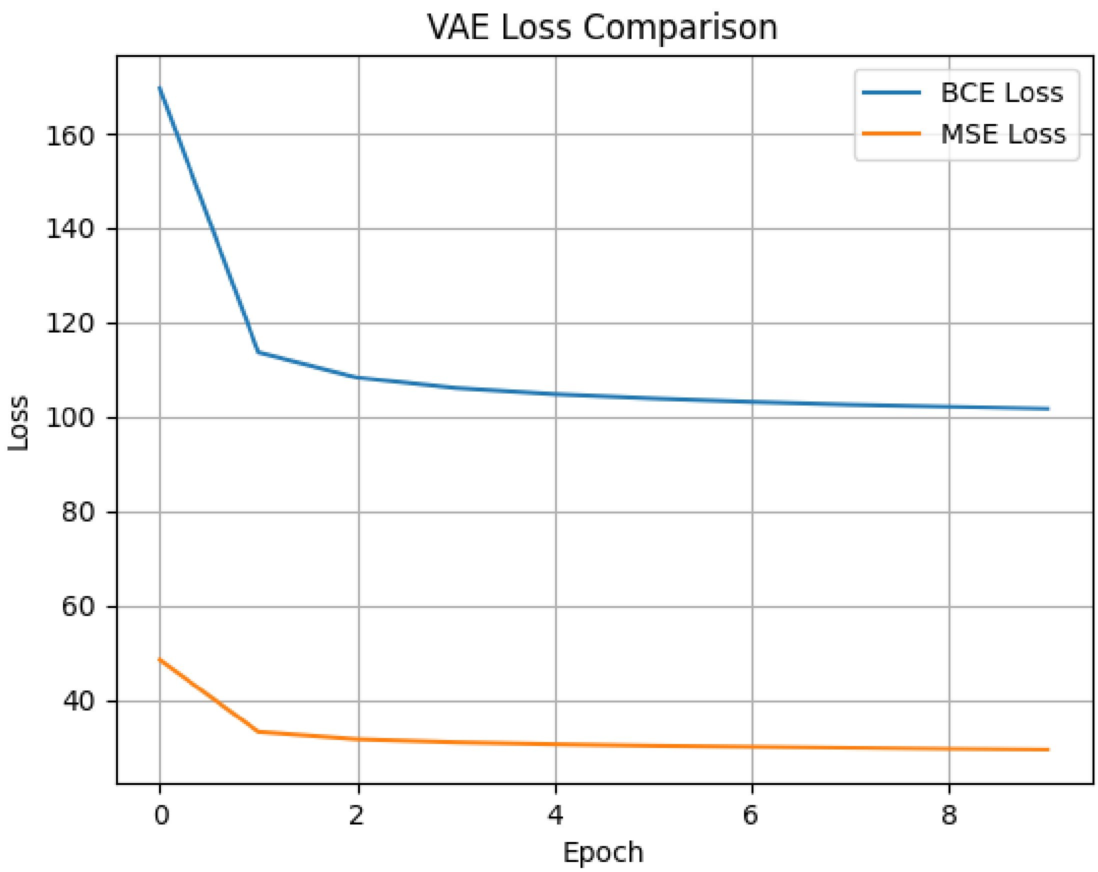


Figure 3: Comparison of reconstruction loss over epochs for VAE models trained with BCE and MSE loss functions.

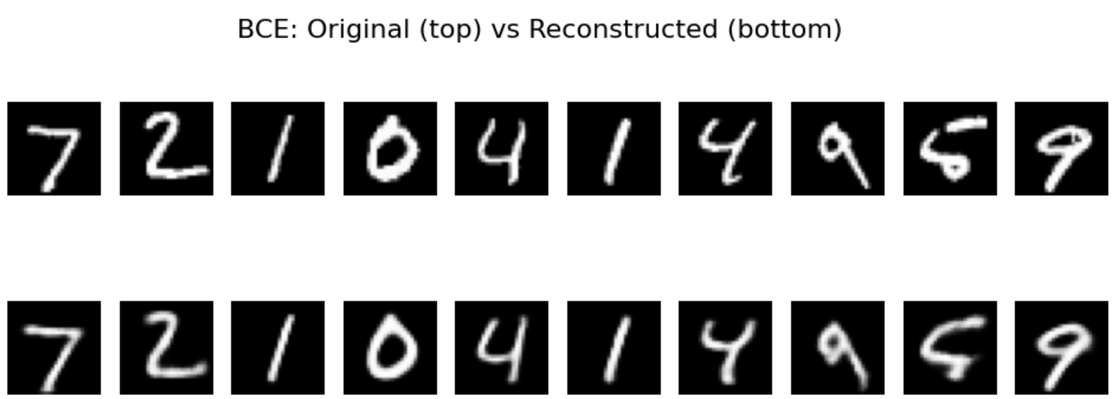


Figure 4: VAE reconstructions trained with BCE loss. Sharp and well-defined digit structures are clearly visible.

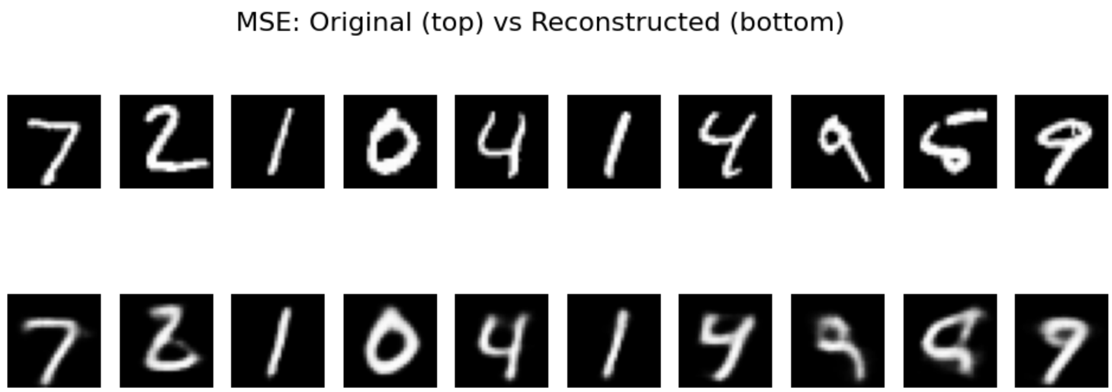


Figure 5: VAE reconstructions trained with MSE loss. Blurred digit outlines are noticeable.

Conclusion: The BCE-based VAE produced clearer reconstructions and converged more quickly, making it more suitable for binary-valued datasets like MNIST. While MSE led to lower numerical loss, the reconstructions lacked sharpness.

Code Link:

Here's the github link of the implemented code: https://github.com/mdsajjadhossain25/CSE4261